



ASSESSMENT:	GROUP ASSIGNMENT	PROGRAMME:	Diploma in ICT
MODULE CODE:	NPRT 630	MARKS:	100
MODULE DESCRIPTION:	Project	DATE:	13 February 2026
EXAMINER(S):	Melvin Kisten	DUE DATE:	TBA (NO EXTENSIONS)
GROUP SIZE:	4 to 6 students		

INSTRUCTIONS:

- Answer ALL questions.
- Indicate your **full names and student number** on your **answer sheet, code, etc.**
- **Copying from a fellow student or any source will be treated as plagiarism.**
- **Submission Requirements**
- **Use Submission template or your submission will not be graded**
- **Report (15 to 20 pages) DOCX and PDF**
- **Video demo**
- Create dedicated GitHub repository, include:
 - Comprehensive codebase Java or Jupyter Notebook or Python or etc scripts
 - Execution instructions
 - Environment configuration
 - Maintain clean, commented code
 - Use version control (**git**):
 - **By forking this repo: https://github.com/login?return_to=%2Fiammelvink%2FNPRT63**
 - Implement error handling

Group Assignment - Project:

Make a working system that:

- Has clean code with Git version control
- Shows error handling
- Includes testing and a clear report
- Ends with a demo

Deliverables:

1. Phase 1 (10%)

- **Title**
- Less than 11 words
- **Introduction**
- Background and context
- etc
- **Problem**
- Specific pain points, affected users
- Avoid vague complaints about "old systems"; identify precise workflow breakdowns, quantifiable inefficiencies
- etc
- **Solution**
- 10-15 concrete functionalities listed
- etc
- **User Experience Improvement**
- Explanation of how system improves usability such as saving time, reducing errors, improving satisfaction, or simplifying tasks
- etc

2. Phase 2 (20%)

- **Business Processes**
- Map each process to user tasks. "Process returns" becomes "customer returns item, cashier processes refund, system updates inventory."
- **Non-functional Requirements**
- Reliability
- Availability
- Security

- Maintainability
- Portability
- Performance
- etc
- **Functional Requirements**
- *Input - Process - Output*
- **UML Diagrams**
- Use case
- Activity
- State
- Class
- Sequence
- etc
- **Work Breakdown Structure (WBS)**
- **Gantt Chart**

3. Phase 3 (20%)

- **Screenshots or mock-ups for each screen, and descriptions**
- Login, registration, authentication screens
- Home, dashboard interfaces
- Data entry forms
- Report generation views
- Admin, user management panels
- **User feedback**
- **Design iteration**
- Revise mock-ups based on user feedback. Show before and after. Explain changes: "Users missed save button, so we made it green and moved it top right."
- **Entity Relationship Diagram**
- Visual representation showing all tables, relationships
- **Entity Descriptions**
- Field name, description, data type, length for each table
- **Relationship Justification**
- Explain one-to-one, one-to-many, many-to-many connections
- **Technology Choice**

- Justify MySQL, Oracle, or other database selection
- etc

4. **Phase 4 (50%)**

- **Working system**
- All features from **Phases 2-3 implemented**
- **Code Quality**
- Clean, well-documented code
- Comprehensive testing performed
- Error handling implemented
- **User testing**
- Include usability testing alongside unit tests. Measure task completion times, error rates, user satisfaction.
- **Demonstration**
- Pre-recorded presentation
- Feature walkthrough
- Technical decisions explained
- Q&A session with supervisor
- **Final Report**
- Consolidates all four phases
- Includes reflection on achievements
- Comprehensive documentation
- Explain your methods
- Show results and discuss them
- List limitations and future improvements
- Reflection - Discuss what you learned about designing for humans. What surprised you? What would you change based on user feedback?

Marking Rubric (100 Marks)

Phase 1

Criterion	What to Look For	Marks	Quick Check Question
Problem Definition	Clear explanation of real user problem based on behaviour, context, and pain points discovered through research.	0-3	“Does report clearly describe who has problem and what difficulty occurs?”
Existing Solutions	Review of current tools, systems, or approaches already attempting to solve problem. Explanation of gaps or weaknesses.	0-2	“Does report explain what already exists and why improvement matters?”
Proposed Solution	Description of system idea addressing identified user problem. Features linked with user needs.	0-2	“Does proposed system clearly solve problem described earlier?”
User Experience Improvement	Explanation of how system improves usability such as saving time, reducing errors, improving satisfaction, or simplifying tasks.	0-2	“Does report explain how user interaction becomes easier or faster?”
Contrast	Clear contrast regarding colours.	0-1	“Does report clearly show colour contrast?”

Performance Levels

Marks	Description
3 Marks	Clear explanation with strong evidence. User context visible. Logical reasoning connects problem, existing systems, and proposed improvement.
2 Marks	Explanation with limited evidence. User context visible. Logical reasoning connects problem, existing systems, and proposed improvement.
1 Mark	Partial explanation. Some context present but weak connection between ideas. No evidence.
0 Marks	Missing explanation or unrelated content. No clear user problem or comparison.

Phase 2

Criterion	What to Look For	Score Range	Quick Check Question
Business Processes	Description explains how users perform tasks using system functions.	1-5	“Can marker follow user workflow step-by-step?”
Functional Requirements	Inputs, outputs, and processing requirements clearly listed.	1-5	“Do listed functions solve user problems?”
Non-Functional Requirements	Performance, usability, security requirements include measurable targets.	1-5	“Are quality targets measurable?”
UML Diagrams	Use case and activity diagrams show interaction between users and system tasks.	1-5	“Do diagrams represent user activity clearly?”

Performance Levels

Criteria	Excellent	Good	Poor
System Analysis & Design (20 Marks)	16-20 Marks - Processes clear. Requirements measurable. Diagrams illustrate real user interaction.	10-15 Marks - Design understandable but missing detail or strong user linkage.	1-9 Marks - Design incomplete or technical only. User perspective absent.

Phase 3

Criterion	What to Look For	Score Range	Quick Check Question
Interface Mock-ups	Screens demonstrate complete user workflow. Design choices explained.	1-5	“Can viewer understand how system works from screens?”
User Feedback	Evidence from testing with real users. Feedback documented. Changes explained.	1-5	“Did users test system and give comments?”
Database Design	ER diagram represents system data requirements.	1-5	“Does database structure match user needs?”
Relationship Justification	Explanation clarifies why relationships exist between entities.	1-5	“Do relationships reflect real user scenarios?”

Performance Levels

Criteria	Excellent	Good	Poor
Prototype & Data Design (20 Marks)	16-20 Marks - Complete prototype, real user feedback, logical database supporting workflows.	10-15 Marks - Prototype functional but explanations limited.	1-9 Marks - Screens missing, no testing, database unclear.

Phase 4

Criterion	What to Look For	Score Range	Quick Check Question
Technology Stack & Implementation	Working system implementing planned features. Code readable and structured.	1-15	“Does system run correctly with understandable code?”
User Testing & Evaluation	Testing conducted with multiple users. Metrics recorded. Improvements applied.	1-15	“Do results show system improvement?”
Demonstration	Video demonstration shows user interaction with system tasks.	1-10	“Can viewer observe system working?”
Final Report	Report documents full project journey with screenshots, testing results, and reflection.	1-10	“Can reader follow entire development process?”

Performance Levels

Criteria	Excellent	Good	Poor
Implementation, Testing & Report (50 Marks)	40-50 Marks - Fully functional system, strong user testing evidence, clear demonstration, structured report.	25-39 Marks - System mostly works. Limited testing. Report adequate.	1-24 Marks - System incomplete. Minimal testing. Report unclear or missing sections.

Mark Allocation Summary

Phase	Marks
Phase 1 - Problem Definition	10
Phase 2 - System Analysis & Design	20
Phase 3 - Prototype & Data Design	20
Phase 4 - Implementation, Testing & Report	50
Total	100 Marks

----- All the best -----